

CONTACT INFORMATION

LOCATION

Colombo, Sri Lanka

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EMAIL

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PORTFOLIO & CODE

github.com/alex-morgan

LINKEDIN

linkedin.com/in/alex-morgan

EDUCATION

B.Sc. (Hons) in Computer Science & Engineering

University of Technology

2023 – 2027 (Expected)

• Focus: Systems Engineering & Networks

• Current GPA: 3.8 / 4.0 (Dean's List)

Diploma in English & Technical Communication

National Institute of Higher Studies

Completed 2023

• Distinction in Professional Writing

PROGRAMMING & TOOLS

Python

Java

C / C++

JavaScript

React.js

Node.js

Linux CLI

Docker

Git / GitHub

Firebase

MySQL

REST APIs

Arduino

Wireshark

CERTIFICATIONS

Networking & Security Fundamentals

Cisco Networking Academy • 2025

Algorithmic Problem Solving

HackerRank Gold Badge • 2024

ALEX R. MORGAN

TECHNICAL SYSTEMS & SOFTWARE ENGINEER

PROFESSIONAL SUMMARY

Solution-oriented and analytical Technical Engineer with a robust background in systems design, algorithms, and full-stack software development. Demonstrated expertise in creating automated test routines, configuring modular IoT frameworks, and architecting database-driven web platforms. Adept at bridging rigorous engineering theory with high-uptime operational execution, maintaining strict code standards, and diagnosing complex technical issues.

KEY ENGINEERING PROJECTS

Algorithmic Sequence Alignment Engine Python 2025

- Engineered an optimized implementation of dynamic programming algorithms (Needleman-Wunsch & Smith-Waterman) for global and local biological sequence alignment.
- Constructed matrix back-tracing routines with dynamic gap penalties, achieving a 35% reduction in execution overhead on large datasets.
- Built comprehensive automated test suites to validate edge cases against published scientific benchmarks.

IoT Multi-Node Telemetry Controller C++ / Arduino / Sensors 2024 – 2025

- Designed and deployed an embedded automation node monitoring ambient temperature, humidity, and relay actuator states in real time.
- Authored non-blocking C++ firmware with circular buffers to ensure zero data packet loss across serial communication channels.
- Integrated automated fail-safe overrides and audio-visual alert triggers for hardware-level safety compliance.

Cloud Asset Management Portal React.js / Firebase / CSS3 2024

- Developed a responsive dashboard with role-based authentication, real-time database synchronizations, and secure file uploads.
- Structured modular React components utilizing custom hooks, cutting component re-render overhead across data tables.

PRACTICAL EXPERIENCE

Technical Solutions Intern Jan 2025 – Present Apex Systems & Cloud Labs

- Created automated Python extraction scripts to parse and correlate internal server log files, reducing diagnostic time by 28%.
- Assisted in routine network audits, monitoring internal switch configurations, and maintaining technical system documentation.
- Collaborated with senior engineers to diagnose and resolve software configuration bugs across staging environments.

ENGINEERING PROFICIENCIES

Methodologies: Object-Oriented Design (OOD), Data Structures & Algorithms, Agile Development, Test-Driven Development (TDD).

Systems: Unix/Linux Environment, Shell Scripting, Hardware Prototyping, Network Protocol Analysis.